

Intraday ETF Performance in the Presence of Surprise Macroeconomic Announcements

Travis Ban

Centre for Applied Financial Studies

School of Commerce, University of South Australia Business School

University of South Australia

Adelaide, 5000, South Australia, Australia

Travis.Ban@mymail.unisa.edu.au

Kartick Gupta

Centre for Applied Financial Studies

School of Commerce, University of South Australia Business School

University of South Australia

Adelaide, 5000, South Australia, Australia

Kartick.Gupta@unisa.edu.au

Petko S. Kalem

Centre for Applied Financial Studies

School of Commerce, University of South Australia Business School

University of South Australia

Adelaide, 5000, South Australia, Australia

Petko.Kalem@unisa.edu.au

Intraday ETF Performance in the Presence of Surprise Macroeconomic Announcements

Abstract

This paper considers the performance of the most popular equity Exchange-Traded Fund (ETF) SPDR S&P 500 (SPY). More specifically, we analyse the SPY ETF's tracking ability of the S&P 500 Index around surprise macroeconomic announcements. We find that surprise macroeconomic announcements increase SPY tracking error during the first 15 seconds after news release. This effect is, however, swiftly removed as SPY returns move back to equilibrium. During the Global Financial Crisis, surprise announcements increased Tracking Error by a factor of 5.5 in comparison to the period before the crisis.

Keywords: Exchange Traded Fund; Intraday Performance; Macroeconomic Announcement; Tracking Error.

JEL Classification: G10; G12; G14

1. Introduction

In this paper we examine the intraday tracking ability of the most popular and highly liquid U.S. equity Exchange-Traded Fund (ETF) SPDR (SPY) of the main equity index - the S&P 500 Index around surprise macroeconomic announcements. More specifically, we analyse the intraday tracking ability of the SPDR by matching real-time U.S. macroeconomic announcement data with tick-by-tick ETF SPY and index values spanning the 10 year period between 2004 and 2013. We ask the following questions: To what extent can Exchange Traded Funds (ETFs) follow their benchmark index? Does the market volatility generated by surprise macroeconomic announcements affect the tracking ability of ETFs and if so, are price deviations created that may be indicative of market inefficiencies? Is ETF performance influenced by the market cycle? The issues explored in the current paper – namely ETF tracking ability of the main index immediately after a surprise news release and how such news surprises affect the intraday ETF performance is of importance to investors, market regulators and stock exchange officials.

The existing ETFs literature suggests that they are able to closely track their benchmark over time, but several uncontrollable factors make perfect replication almost impossible. Although extant literature investigates the performance of ETF using a longer frequency data (e.g. daily or monthly), there is relatively scant evidence on the performance of ETF tracking ability using intraday data (Elton et al., 2002). Such an analysis is important for a number of reasons.

First, ETFs are rapidly increasing as an investment tool as they are a fast and convenient way to replicate almost any available index. ETFs were introduced to the U.S. market in 1993 with the creation of the Standard & Poor Depository Receipt (SPDR) and have since experienced rapid, global growth. For instance, at the end of 2003 there were just over \$151 billion in Assets under Management (AUM) held in U.S. ETFs (Investment Company Institute,

2014). Twelve years later, the AUM in EFT experienced an exponential increase to reach about \$2.932 trillion by the end of June 2015 (Black Rock, 2015). These investment vehicles aim to follow an index by either directly purchasing a representative pool of securities or synthetically swapping for the underlying index return. Gastineau (2001) suggests ETFs offer many advantages, triggering recent growth with increasing volume and liquidity, which does not appear to be slowing down. There are currently more than 2,090 ETFs in U.S. only and about 5,490 ETFs globally with the ability to track almost any index or benchmark around the world (p. 6, Black Rock, 2015).

Second, both computerized trading and its subgroup of superfast trading - High Frequency Trading are increasing exponentially over the last several years, so a more detailed understanding of short-term ETF movements is required.¹ Goldstein et al. (2014) survey several AT and HFT studies focusing on the evolution of automated trading. More recently, O'Hara (2015) considers the market microstructure changes in light of computerized trading technology.

Third, macroeconomic announcements impact the underlying index. Considering that the core objective of an ETF is to follow a particular index, macroeconomic announcements will also influence ETF (mis)pricing.

Finally, we study the evolution of ETF performance during times of substantial market changes by examining periods before, during and after the most recent and devastating global crisis – the Global Financial Crisis (GFC) of 2007-2008. More specifically, by matching real-time macroeconomic announcement data with tick-by-tick ETF and index values spanning the period between 1st January 2004 and 31st December 2013, we analyse intraday tracking ability of the most liquid equity ETF – the SPDR S&P 500 ETF (SPY) to the main stock market index

¹ For detailed definition of the HFT and review of the empirical HFT literature in US equity markets see SEC (2014).

– S&P 500. The announcement sample includes 1,104 observations spanning 12 different economic indicator types that announce new information during the NYSE trading hours. By using a 10 year sample period, intraday ETF performance during different stages of the market cycle can be compared, as well as its time-varying nature.

Our empirical findings suggest that surprise macroeconomic announcements reduce the tracking ability of ETFs immediately after a news release by increasing the deviation between ETF and index returns. However, this surprise effect is only evident during the first 15 seconds after each announcement. While analysing how surprise news affects performance during different stages of the market cycle, we find that a larger tracking error around surprise announcements is evident during a market downturn compared to a growth period. This occurs because investors are more influenced by unexpected economic changes during a time of high market negativity and volatility. The empirical findings are robust for several different announcement types made during the trading day and over a broad timeline.

We add to the existing literature in several ways. First, these findings are essential to regulators and practitioners who must be vigilant when releasing an official announcement which is considerably different to its expected value. On 6th May 2010, many U.S. equity products experienced an extremely rapid decline and recovery. Several ETFs declined by over 60 percent while deviating from their underlying index by 15 percent or more (Securities Exchange Commission, 2010). To safeguard against another Flash Crash, the Securities and Exchange Commission (SEC) and the Commodity Futures Trading Commission (CFTC) published the Flash Crash Report, detailing the source of the event and highlighting several preventative measures (Securities Exchange Commission, 2010). Therefore, this event highlights the vulnerability of ETFs and establishes the importance of intraday ETF movements.

Second, we examine the intraday relationship between surprise U.S. macroeconomic announcements and the tracking ability of a highly liquid U.S. ETF using high-frequency, tick-by-tick data. It details a factor that alters ETF tracking error during trading hours, an aspect that has been previously overlooked.

Finally, this study increases the understanding of ETFs and establishes the importance of surprise macroeconomic announcements by considering the determinants affecting ETF performance. Investors will require an understanding of intraday ETF movements as they continue to grow as an investment tool, as transaction costs decrease, and as high-frequency trading increases. It highlights a moment during the trading day when ETFs deviate from their ‘true value’ which is vital for investors attempting to time the market or profit from small market movements. These findings will also be of interest to macroeconomic analysts whose forecasts can directly influence ETF performance; and to academic researchers as this study establishes a benchmark for further analysis of the short-term factors affecting ETF performance.

The paper proceeds as follows. The relevant literature is reviewed in Section 2. The data utilized in our study are presented in Section 3. The results are reported in Section 4. Section 5 concludes.

2. Literature Review and Hypotheses Development

Research into ETF performance and efficiency mostly examines the factors that affect the medium to long-term tracking ability of ETFs using low frequency (daily) data (see among others, Aber et al., 2009; Ackert and Tian, 2008; Drenovak et al. 2012; Elton et al., 2002; Kostovetsky, 2003; Qadan and Yagil, 2012; Rompotis, 2005, 2007). The existing literature proposes that ETF performance can be measured in a similar way to passive mutual funds as they share many similar characteristics. As with passive mutual funds, ETFs attempt to follow

an index, so a perfectly managed fund would synchronously move in the same direction as the market, at the same time and by the same degree. Performance is most often determined by analysing the funds premium or discount compared to the underlying index or with tracking error. In this case, positive and negative deviations are treated equally because any deviation is undesired, given the goal of the fund is to track an index.

In theory, ETFs should perfectly follow index movements by fully replicating the underlying benchmark or synthetically trading for the index return. Underlying equity values are easily calculable so investors monitor the Net Asset Value of each ETF compared to its traded value, and ETF managers are able to create and redeem securities to ensure ETF value is close to its Net Asset Value. Therefore, ETF prices should be able to follow precisely the underlying benchmark value if its assets under management are a perfectly representative basket.

In reality however, several factors affect ETF performance and make perfect tracking almost impossible. Chiang (1998), Frino and Gallagher (2001) and Keim (1999) suggest that transaction costs, fund cash flows, dividend treatment by the index, benchmark volatility, index liquidity, corporate activity and index composition changes will factor into tracking error creation. Considering each index is calculated as a ‘paper’ portfolio where transactions can occur instantly, in unlimited quantities and without cost, index composition can vary without constraint (Perold, 1988).

Elton et al. (2002) document that management fees, transaction costs and the chosen replication strategy lead to temporary diversion and imperfect performance. The authors show that deviations disappear within one day. Elton et al. attribute this to arbitrage opportunities and further suggest the pricing pressure associated with arbitrage should cause deviations to disappear. Building on Elton et al. (2002), Marshall et al. (2013) examine high-frequency data for U.S. ETFs and test whether arbitragers are able to profit from the intraday price deviations.

Marshall et al. argue that investors who buy (sell) an ETF when it is under- (over-)priced, would be able to obtain net returns of 6.7 percent above the market rate.

Frino and Gallagher (2001) examine the tracking performance of 42 S&P 500 index funds. The authors find that the magnitude of tracking error is directly related to the level of underlying index volatility. The scholarly literature on the factors that fuel in the tracking error over the long-run suggests that ETFs are highly influenced by volatility induced by the underlying benchmark index. Engle and Sarkar (2006) study U.S. and International ETFs and find that price deviations are positively related to volatility in the ETFs underlying Net Asset Value. These results are supported by Qadan and Yagil (2012), who examine U.S. ETFs' daily returns between January 2006 and December 2008.

Although no direct analysis has been conducted on how surprise macroeconomic announcements influence ETF tracking error, some studies examine their influence on ETF pricing. Andersen et al. (2007) use a high frequency dataset to examine global stock, bond and foreign exchange markets and find that unexpected news produces conditional mean price jumps. A repositioning effect is also evident when unexpected news is released, stemming from information asymmetry between market participants. Balduzzi et al. (2001) argue that volatility persists for a prolonged period as investors have varying degrees of skill interpreting the new information. Brock and LeBaron (1996) suggest persistence is due to lower levels of transparency and information precision.

Birz and Lott Jr (2011) and Brenner et al. (2009) analyse how U.S. equity market volatility responds to surprise macroeconomic announcements and report that a volatility increase is directly related to the release of unexpected macroeconomic news. Brenner et al. (2009) examine stock market returns around four different economic indicator types from 1986 to 2002 and argue that a surprise news release generates a persistent volatility increase. This increase in volatility can last from several minutes to a number of hours, depending on the level

of liquidity within the market (Nowak et al., 2011). Therefore, unexpected news may quickly change the price of assets in the underlying equity index, but it will also substantially increase the volatility of the underlying securities (Brenner et al., 2009; Kim and Verrecchia, 1991).

Chung and Hrazdil (2013) analyse the speed of convergence to market efficiency within the ETF market. The authors investigate intraday data for ETFs between January and July of 2008, and find the market fully adapts to new information within 30 minutes. This is consistent with the study by Wang et al. (2006), who find that surprise central bank communications increase the volatility of ETF prices. These results suggest ETFs follow the base (spot or synthetic) market by increasing volatility around announcements.

In summary the extant literature has established the following two: (i) an increase in market volatility increases ETF's tracking error; (ii) an unexpected macroeconomic announcement will generate higher market volatility. Therefore, ETF performance should decrease after a surprise macroeconomic announcement, with high surprise announcements generating a larger deviation between ETF and index returns, compared to lower surprise announcements. Hence the following hypothesis is stated in an alternative form:²

H₁: SPY tracking error around high surprise macroeconomic announcements will be significantly higher, ceteris paribus.

Andersen et al. (2007) argue that highly liquid, efficient markets are able to factor new information into stock prices rapidly. However, Brenner et al. (2009) finds that volatility within the underlying market will remain at an increased level for several minutes after a surprise macroeconomic announcement, depending on market liquidity. This is due to investors differing ability to process information (Kim and Verrecchia, 1997). In theory, ETF tracking error is strongly influenced by market volatility, and a volatility increase associated with

² All hypotheses are stated in an alternative form.

surprise news will persist, so any tracking error increase generated by a surprise announcement should stay persistently higher. Therefore, the second hypothesis developed in this study is:

H₂: A high surprise announcement will significantly increase ETF tracking error during time periods after the conclusion of the initial 15 second interval, *ceteris paribus*.

The third hypothesis addresses how tracking error changes over-time and through changes in the market cycle. Qadan and Yagil (2012) examine long-term ETF tracking error and find it is substantially higher during a period of high market uncertainty (GFC). The authors attribute this occurrence to increased volatility and investor irrationality, as ETFs deviate further from their ‘true value’ when market sentiment is negative and large price swings are common. Therefore, the third hypothesis in this study is:

H₃: Surprise macroeconomic announcements will affect tracking error differently immediately after the announcement, depending on the stage in the market cycle, *ceteris paribus*.

3. Data and Methodology

Data

Data for both the SPY and S&P 500 between 1st January 2004 and 31st December 2013 are provided by Securities Industry Research Centre of Asia-Pacific (SIRCA) from Thomson Reuters Tick History (TRTH). This database records tick-by-tick observations and is time-stamped to the nearest millisecond, with several ETF trades occurring each second, on average. However, the S&P 500 value is recorded every 15 seconds until December 2011, and every five seconds from January 2012 until December 2013. Following Engle and Sarkar (2006) and Tse and Martinez (2007), a dataset of SPY prices is generated using the mid quote for each one second interval. To generate index prices at one second intervals, the most recently reported index value is carried forward until a new value is reported.

Announcement data are time-stamped to the millisecond and is gathered through the SIRCA. Each announcement must occur during the NYSE Arca trading day so relevant return data is available at high-frequency intervals for both the SPY and S&P 500. Any announcement occurring before 9:40am or after 3:20pm is excluded to minimize contamination from opening or closing trades. In addition, a recent forecast for each economic indicator must be available to determine the level of surprise in the official announcement. All official announcements and forecasts that meet the study requirements between 1st January 2004 and 31st December 2013 are included within the sample, as in Table 1.

[Insert Table 1 here]

Building on Andersen et al. (2007), we include 25 economic indicator types that refer to regular scheduled announcements. We remove 1,283 observations that are announced occur prior to 9:40am and 64 observations take place after 3:20pm. A further 86 observations are removed as no forecast information is available, leaving 1,104 individual announcements. 12 of the original 25 indicator types meet the requirements necessary for this study. Nine of the 12 economic indicators have at least 104 announcement observations spread across the 10 year sample with monthly release dates. However, the Durable Goods and Target Federal Funds Rate indicators have only 10 and 13 observations respectively, while Business Inventories reports 88 observations. This is due to a release time change for Durable Goods (8:30am), a lack of change in the Federal Funds Rate for the Target Federal Funds Rate indicator, and announcements without a reported forecast for Business Inventories.

Considering most of the 12 economic indicators are reported monthly, announcement dataset distribution is spread somewhat evenly between each year and month from 1st January 2004 until 31st December 2013. However, the majority of announcements are reported on a Tuesday and a Thursday, while nine economic indicators are released at or just after 10:00am

each day. Consumer Credit is announced at 3:00pm, Target Federal Funds Rate is announced at 2:15pm and Federal Budget Balance is reported at 2:00pm, on average.

An official announcement is defined as the first public mention of an official value reported for a specific economic indicator. Several forecasts are made between each official announcement, generating a market expectation for each economic indicator. The forecast released closest to an official announcement release is used as its expected value when examining announcement surprise.

To determine the level of surprise within each announcement, we follow Andersen et al. (2003), Andersen et al. (2007), Balduzzi et al. (2001) and Green (2004) and construct a standardized measure. This measure, known as *Standardised News*, allows for a meaningful comparison of different economic indicators regardless of the unit of measurement, capturing only the level of surprise within each announcement. For example, a *Standardised News* value of 2.50 indicates that the reported value of the announcement is 2.5 standard deviations away from the expected value, suggesting a high level of surprise. The difference between the actual value announced and the expected value is calculated to create a ‘surprise’ value, before dividing it by the standard deviation of all surprises for a single indicator type during each year.

As the scope of this study is not to study the effect of positive or negative news on the ETF performance, we use absolute *Standardised News*. Whether an announcement is surprisingly excellent or surprisingly poor is insignificant because the relationship between ‘surprise’ and ETF performance is of importance instead of how different announcements affect performance.

Next, the announcement date and time are matched to this dataset to gather the price of the ETF and index every second for 30 minutes before and after each announcement. This generates a dataset with the value of the ETF and index before and after each announcement occurs. Returns for the ETF and index are calculated following Andersen et al. (2007) and Chung and Hrazdil (2013) who use the natural logarithm of price change each period. As S&P

500 values are reported every 15 seconds until 2011, the smallest gap used between return time intervals is 15 seconds for both the ETF and index to maintain consistency. By doing this, returns with a zero value that arise solely due to a lack of index information are minimised.

To examine ETF performance, short-term, intraday ETF tracking error is measured. This measure is defined as absolute return difference between the return of the ETF at time t , and the return of the benchmark index at time t . By applying this equation to SPY and S&P 500 returns, SPY tracking error is generated for each 15 second interval before and after every announcement. However, tracking error during the first 15 seconds after each announcement is used as the main dependant variable because the biggest increase in market volatility and tracking error is expected during this time.

Empirical Model

To capture how SPY tracking error is affected by surprise macroeconomic announcements, a panel regression is employed where SPY tracking error is regressed on the level of surprise (*Standardised News*) corresponding to each announcement. Several of the factors that are known to influence tracking error are incorporated as control variables in Equation 1:

$$TE_t = \beta_0 + \beta_1 SN + \beta_2 VOLA + \beta_3 AVEQS + \beta_4 TDIF + \beta_5 MANN + \beta_6 INDIC + \beta_7 HOUR + \beta_8 DAY + \beta_9 MONTH + \beta_{10} YEAR + \varepsilon, \quad (1)$$

where TE_t is the absolute deviation between SPY and S&P 500 returns, SN is the absolute *Standardised News* for each announcement, $VOLA$ is a volatility proxy measured as the standard deviation of 1 second ETF's returns during the 10 minutes prior to each announcement, $AVEQS$ is the average quoted bid-ask spread – a proxy for liquidity, $TDIF$ measures how 'old' the forecast of an announcement is, $MANN$ examines a potential contamination effect due to multiple announcements, and $INDIC$, $HOUR$, DAY , $MONTH$, and $YEAR$ control for the type and time of the announcement.

Although an index volatility increase associated with a surprise macroeconomic announcement is crucial for this study, any risk not associated with announcements must be controlled for as it is a major source of ETF tracking error (Frino and Gallagher, 2001; Qadan and Yagil, 2012). To this end, the standard deviation of one second ETF returns for the 10 minutes prior to the announcement is used following Rompotis (2007). One second returns leading up to each announcement are used to ensure the volatility experienced in the market is captured while excluding volatility associated with surprise.

Ackert and Tian (2008) find that tracking errors decrease with higher liquidity. Therefore increased liquidity implies more investors, more competition, and hence, less arbitrage opportunities in the market. A liquidity measure is included in the regression to ensure any changes in SPY liquidity over the 10 year sample period are accounted for. The average quoted bid-ask spread is measured for the SPY at one second frequency over the 10 minutes prior to each announcement. The average is used as a liquidity measure as it measures the spread evident in the market at the time of the news release (Stoll, 1989). The total number of units traded over the 10 minutes prior to the announcement is also considered (TVOL) as a control variable. This is to account for changing volume levels, as increased volume is associated with decreased tracking error, relative to low volume periods (Ackert and Tian, 2008).

When an announcement is made, there can often be conflicting news released at the same time or relatively close to the announcement in question. This could generate conflicting results as the ETF tracking error could change due to the release of one announcement but the regression could associate it with the other. Therefore, a multiple announcements dummy variable is included (*MANN*) to control for this event. ‘Contaminated announcements’ are defined as events with one or more other news releases occur within half an hour before or after the announcement in question.

Not all announcements affect the index or the SPY by the same degree (Andersen et al., 2007; Birz and Lott Jr, 2011). To control this 12 dummy variables representing each economic indicator type are included (*INDIC*) to capture the effect of each on tracking error. The existing literature suggests that both index volatility and ETF tracking error fluctuates throughout the day (Engle and Sarkar, 2006), during the week, during the year (Rompotis, 2007), or over the market cycle (Qadan and Yagil, 2012). Therefore, we use dummy variables for the hour of the day (*HOURL*), day of the week (*DAY*), month (*MONTH*) and year (*YEAR*) are used to capture time-varying affects.

4. Results

4.1. Preliminary Data Analysis

To understand SPY tracking error and its intraday movements, a baseline must be obtained before factoring in surprise. This is undertaken by comparing SPY and S&P 500 returns, illustrated in Figure 1. This graph details the average 15 second tracking error from 5 minutes before each announcement until 10 minutes after its release. As depicted in Figure 1, a spike in tracking error is evident directly after each announcement. This suggests the SPY is unable to perfectly follow the S&P 500 as an announcement increase ETF tracking error. However, these results do not consider surprise announcements.

All 12 economic indicators chosen for this study influence the broad U.S. economy, but each reports substantially different announcement values. For example, the average announced value (unreported) for Construction Spending is -0.04 with a standard deviation of -0.92 while New Home Sales announcements average a value of 11 with a standard deviation of 112. Also, each economic indicator exhibits different levels of average surprise. This is most evident while analysing the raw surprise of each announcement, measured as the actual announced value for each news release minus its forecast. For instance, the average surprise value for Purchasing

Managers Index is 0.40 with a standard deviation of 2.10, while Consumer Credit has an average surprise of 0.13 and a standard deviation of 6.24 (unreported). As each indicator type is significantly different, announcements can be compared after standardisation. Figure 2 outlines the distribution of raw *Standardised News* values associated with each announcement. Here, 836 macroeconomic announcements have a *Standardised News* value between -1 and 1, while only 81 announcements have a *Standardised News* value below -2 or above 2. As expected, most announcements have a near zero value because most announcements are not a surprise to the market. However, we use absolute *Standardised News* values rather than raw values to ensure any surprise, whether positive or negative, can be captured. The further an announcement is from zero, the less likely market participants expect it.

Table 2 reports the descriptive statistics of each control variable. The table shows that absolute *Standardised News* is skewed, with most observations occurring at the lower end. Also, the average announcement records its forecast 2.2 days before the release of an official value (TDIF), but some announcements release a forecast up to one month prior to its official release. Table 2 does not include descriptive statistics for the volume variable (TVOL) as a high correlation between SPY volume and volatility (VOLA) requires its removal from the analysis.

[Insert Table 2 here]

Table 3 shows the correlation matrix, finding that TVOL and VOLA generate a correlation coefficient of 0.7228. The highest correlation coefficient found within the remaining variables is 0.2874 between VOLA and AVEQS. Nevertheless, in unreported results we find that our baselines regressions remain strong even after including TVOL is included as an additional control variable.

[Insert Table 3 here]

4.2. Initial Investigation

Model 1 of Table 4 reports an initial investigation without control variables. This model regresses the SPY tracking error obtained during the first 15 seconds after each announcement against the absolute *Standardised News* value for each announcement, producing a coefficient of 0.00014 for absolute *Standardised News* and statistically significant at the one percent level. This result rejects the null hypothesis for H_1 ; suggesting surprise macroeconomic announcements increase ETF tracking error. However, this regression excludes variables that are important determinants of tracking error.

In Model 2 of Table 4, control variables are added to the regression while excluding fixed effects. Model 3 excludes control variables while including fixed effect variables. In both scenarios, absolute *Standardised News* is positive and statistically significant at the one percent level, regardless of which variables are included in the analysis. By including the control variables in Model 2, there is substantial and statistically significant improvement of the explanatory power of the empirical model with controls which results in a reported adjusted R^2 of 26.05% - over the simple linear regression model (Model 1) with an adjusted R^2 of 1.47%. In addition, the volatility significantly increases ETF tracking error as expected but the liquidity variable is statistically not significant. This model also suggests a longer time between a forecast and an announcement (TDIF) and multiple announcements effect (MANN) significantly affect tracking error.

In Model 4 of Table 4 we include all control variables and fixed effects. The absolute *Standardised News* coefficient is 0.00012, statistically significant at the one percent level. Therefore, a one point increase in macroeconomic announcement surprise is expected to generate a 0.00012 increase in tracking error. Considering the mean 15 second SPY tracking error from five minutes before until 10 minutes after each announcement is 0.00020 (unreported), this increase generated by a surprise macroeconomic announcement is

substantial. For instance, if announcement n is released with a *Standardised News* value of 3.0, the expected tracking error during the 15 seconds after the announcement would be approximately 0.00056. Consequently, the null hypothesis for H_1 is rejected as an increased level of surprise in announcement n will generate an increased tracking error for the SPY, *ceteris paribus*. These results have important implications as they suggest that surprise announcements play an important role in tracking error determination.

Based on the findings of existing studies, a decrease in liquidity or an increase in volatility is expected to increase tracking error, *ceteris paribus* (Frino and Gallagher, 2001; Keim, 1999). We provide new insights on intraday SPY performance by finding that an increase in the quoted bid-ask spread (decrease in liquidity), or an increase in the standard deviation of one second returns during the 10 minutes prior to each announcement (increase in volatility) is significantly associated with an increase in SPY tracking error. Contrary to expectation, the time difference between a forecast and an official announcement release (TDIF) is unlikely to influence tracking error. Also, multiple announcements dummy (MANN) is statistically insignificant. We further pursue the effect of multiple announcement on tracking error in Section 4.5 by splitting the sample into contaminated and uncontaminated observations.

[Insert Table 4 here]

4.3. Tracking Error Deviation Persistence

Although tracking error is significantly increased by an increase in macroeconomic announcement surprise during the first 15 seconds after a news release, this analysis suggests the effect is rapidly removed. Table 5 examines how tracking error is influenced by ‘surprise’ during the second (15, 30), third (30, 45) and fourth (45, 60) 15 second time intervals after each announcement. In each scenario, all control variables and fixed effects are included in the regression. Although the coefficient for absolute *Standardised News* is positive, it is insignificant in each case. Therefore, the null hypothesis for H_2 cannot be rejected as the

tracking Error change generated by a surprise announcement does not persist for more than 15 seconds. This suggest that the SPY is highly efficient and is able to remove any deviation generated by unexpected announcements within the first 15 seconds.

An exception can be found when examining the tracking error over the entire first minute after the announcement (0, 60), where *Standardised News* is positive and statistically significant at the one percent level. Although tracking error for high *Standardised News* announcements during the last 15 seconds of this period – the period between (45,60) – is not statistically different to low *Standardised News* announcements, the deviation in returns obtained during the first 15 seconds is high enough to create a significant result when grouping the full 60 seconds together.

In the 6th (-15, -1) and 7th (-30, -15) Columns of Table 5, the tracking error prior to each announcement is examined at 15 second intervals. In theory, the absolute *Standardised News* value should be insignificant as unexpected news is yet to be released to the market. In this case, investors would still be expecting the forecasted value. This expectation is obtained two time periods before each announcement (30 seconds prior until 15 seconds prior). However, absolute *Standardised News* is statistically significant at a one percent level before each announcement. A one point increase in *Standardised News* is associated with a 0.00006 increase in SPY tracking error prior to each announcement. These results suggest surprise announcements may be anticipated before they are released. This can be attributed to trading based on an expectation of a different result in the seconds leading up to a surprise announcement.

Although tracking error is significantly influenced by a surprise announcement during the 15 seconds leading up to the event, the difference between high *Standardised News* and low *Standardised News* is smaller than during the period after the announcement. This is represented by a coefficient that is substantially lower (0.00006) than the one associated with

the 15 seconds after a news release (0.00012). John and Lang (1991) argue that a small lead-up in price and volatility prior to an announcement release is evident, and the results of this study suggest this could be apparent in the ETF market.

In the time periods leading up to, and after each announcement, most control variable results do not significantly change. However, multiple announcements dummy (*MANN*) is statistically significant when examining either the first full minute after each announcement, or during the second, 15 second time interval prior to each announcement ($-30, -15$). This could be due to several contaminating variables being spread out over the initial minute after each announcement, or just prior to some announcements. This is further examined as a robustness check.

[Insert Table 5 here]

4.4. Sub-Period Analysis

There is a strong relationship between long-term market volatility and ETF tracking error, and tracking error can be substantially different from year to year. Therefore, the announcement sample is split into three distinct sub-periods to examine how surprise announcements influence SPY tracking error during different stages of the market cycle.

Qadan and Yagil (2012) examine long-term tracking error leading up to the Global Financial Crisis (GFC) and during the GFC, finding average tracking error is substantially higher during the crisis. They associate this with higher market volatility and argue that investor sentiment is negative and the market is highly volatile during this time. The SPY tracking error is at its peak during 2008 and 2009, coinciding with the GFC (Figure 3). This supports the theory that tracking error is influenced by long-term market volatility, but it does not examine the role of surprise announcements each year.

A growth period is analysed in Model 1 of Table 6 (2004-2006), capturing all announcements between 1st January 2004 and 31st December 2006. The study finds that

absolute *Standardised News* is positive and statistically significant at the five percent level, generating a coefficient estimation of 0.00003. This value is substantially smaller than the one obtained while examining the full 10 year sample (0.00012), supporting the view that surprise announcements influence SPY tracking error by a smaller degree during a growth period.

The second sub-period spans 1st January 2007 to 31st December 2009 to examine how SPY tracking error responds to surprise news during a market downturn, with the results shown in Model 2 (2007-2009) of Table 6. Here, absolute *Standardised News* is positive and statistically significant, producing a coefficient estimation of 0.00018. A one point increase in absolute *Standardised News* is expected to increase SPY tracking error by 0.00018, *ceteris paribus*. This value is above the 10 year average and substantially higher than the 2004 to 2006 period. Considering overall market volatility is controlled for, this is a notable result because a surprise announcement during a time of overall market fear and uncertainty (2007-2009) is expected to increase SPY tracking error by a factor of 5.5 compared to the same announcement during a growth period (2004-2006). Therefore, the null hypothesis for H_3 is rejected, as tracking error is substantially higher during a market downturn compared to a market growth period.

As a tracking error increase can be generated by the inequality of investor's ability to process new information at the same rate, this effect is compelling as investors are more concerned with possible economic changes that arise with a surprise macroeconomic announcement when there is widespread, negative sentiment (Kim and Verrecchia, 1997). An announcement with a high level of surprise can shock investors, generating substantial volatility and a higher tracking error, *ceteris paribus*. In a positive market, investors may respond in a more reasonable manner, increasing tracking error by a smaller degree.

Model 3 (2010-2013) reports the results obtained by analysing tracking error between 1st January 2010 and 31st December 2013. The absolute *Standardised News* coefficient of 0.00012 is statistically significant, as is the case in each sub-period. Considering a more volatile market

increases the effect surprise announcements have on tracking error, the coefficient for absolute *Standardised News* between 2010 and 2013 is expected to be higher than the 2004 to 2006 period but lower than the 2007 to 2009 because the market is relatively stable during this time. A coefficient estimation of 0.00012 is higher than the first period (0.00003) and lower than the second period (0.00018), providing evidence of this expectation.

Although multiple announcements (MANN) and the time difference between a forecast and an official announcement (TDIF) are statistically insignificant while determining tracking error during each period, the volatility (VOLA) and liquidity (AVEQS) measures fluctuate between periods. SPY volatility and liquidity is insignificant between 1st January 2004 and 31st December 2006 while both significantly influence SPY tracking error between 2007 and 2013.

[Insert Table 6 here]

4.5. Robustness Checks

When residuals are correlated across observations, OLS standard errors can be biased which can lead to either overstated or understated coefficient estimates (Petersen, 2009). Considering observations could be correlated within each announcement type, we follow Rogers (1994) by conducting a clustered standard error test to determine whether indicator type correlation influences the results. This regression is reported in Model 1 of Table 7. Absolute *Standardised News* is still statistically significant at the one percent level, implying that indicator type correlation is unlikely to influence the results.

In a similar note, observations can correlate by both announcement type and time period. Therefore, the two dimensional clustered standard error approach outlined by Petersen (2009) is applied to cluster by both variables. Here, tracking error is clustered by indicator type and year, reported in Model 2 of Table 7. These results are similar to the results found when using a single cluster approach (Model 1 of Table 4) and an approach that does not account for clustering (Model 4 of Table 4). Although the time difference between a forecast and an official

announcement (TDIF) is significant, all other results remain the same. Therefore, correlation within indicator type and year is unlikely to influence the effect surprise macroeconomic announcements on SPY tracking error.

An investigation of SPY tracking error during the first 15 seconds after each announcement suggests multiple announcements dummy is not significant (Table 4). However, this variable is statistically significant while analysing the tracking error obtained in the first 60 seconds after each announcement and during the 15 second interval beginning 30 seconds prior to each announcement (Table 5). As these results contradict, further analysis is necessary. In Model 3 and 4 of Table 7, the announcement sample is split into uncontaminated and contaminated observations respectively. In this instance, *Standardised News* is positive and statistically significant for uncontaminated announcements. A one point increase in absolute *Standardised News* for an announcement that is released at least half an hour away from another announcement is expected to increase tracking error by 0.00016. This is a notable result, and it is higher than the coefficient estimation found in the original regression (0.00012). However, absolute *Standardised News* is statistically insignificant while examining the contaminated announcement sample (Model 4). This suggests that, although the surprise component in an announcement does not significantly influence SPY tracking error when another announcement is made within half an hour of the announcement in question, surprise macroeconomic announcements will impact tracking error to a higher degree when an announcement is made free of any multiple announcements.

This could be explained by the level of surprise within each announcement. Balduzzi et al. (2001) argue that it can be difficult to separate the effects associated with each announcement observation when different announcements are released concurrently. Several multiple announcements with a high absolute *Standardised News* value could occur at approximately the same time as a low surprise announcement, generating noise and hiding the impact of

‘surprise’. Therefore, the multiple announcement effect may play a factor in affecting the results, as a substantial tracking error increase associated with a high absolute *Standardised News* value may incorrectly be linked to a low *Standardised News* announcement released at the same time. However, this effect is not strong enough to significantly alter tracking error when examining all announcements between 1st January 2004 and 31st December 2013.

[Insert Table 7 here]

In summary, surprise macroeconomic announcements are associated with an increase in SPY tracking error during the first 15 seconds after it is reported, and this effect is stronger during poor market conditions. However, the deviation generated by a surprise announcement is not persistent as SPY tracking error returns to a sustainable level after 15 seconds. These results are robust when clustering standard errors, but contaminated announcements (one or more announcements made within half an hour of each other) influence the strength of the surprise announcement effect. Therefore, an uncontaminated high surprise macroeconomic announcement will increase SPY tracking error immediately after its release.

5. Conclusion

By analysing the intraday pricing relationship between the SPY and the S&P 500, we deliver insights into ETF tracking error fluctuations around surprise macroeconomic announcements. High frequency data between 1st January 2004 and 31st December 2013 is utilised to examine how SPY tracking error responds to varying levels of surprise found within different types of announcements. Time-stamped, macroeconomic announcements with a forecast made during trading hours are used, and surprise is standardised to ensure each different indicator type is comparable.

We find that surprise macroeconomic announcements reduce the tracking ability of the SPY for a short period after each announcement. The market adapts quickly to new information,

removing the surprise premium from tracking error within 15 seconds. This effect is evident prior to each announcement as tracking error slightly differs between high and low surprise announcements during the 15 seconds leading up to each news release.

The surprise effect fluctuates depending on the stage of the market cycle. Surprise macroeconomic announcements generate a significantly larger deviation during a recession compared to the same announcement in a stable or growth period, *ceteris paribus*. Investors become more irrational during a time of market uncertainty, leading the ETF to deviate by a larger degree before the overall market adapts to new information.

This paper adds to the existing literature as existing studies do not consider the intraday factors affecting ETF performance. To the best of our knowledge, this is the first study to document a strong linkage between macroeconomic announcements and tracking error in an intraday setting. Until now, most studies concentrate on the factors affecting long-term performance, but as ETFs grow as an investment tool, transaction costs decrease, and high-frequency trading increases, an analysis of the factors affecting intraday performance is necessary.

During the Flash Crash on 6th May 2010, a substantial number of ETFs were among the securities hardest hit by sharp price declines. Institutional investors became highly irrational during this time of market unrest and used ETFs as a tool to reduce market exposure. As some U.S. ETFs deviated from their benchmark indices by more than 15 percent during this time, regulators realised the need for a complete understanding of the factors affecting intraday ETF performance to ensure another crash does not occur. Therefore, these findings have strong implications for regulators, who should take care when announcing unexpected macroeconomic information during a time of market unrest.

REFERENCE

- Aber, J. W., Li, D., & Can, L. (2009). Price volatility and tracking ability of ETFs. *Journal of Asset Management*, 10(4), 210-221.
- Ackert, L. F., & Tian, Y. S. (2008). Arbitrage, liquidity, and the valuation of exchange traded funds. *Financial Markets, Institutions & Instruments*, 17(5), 331-362.
- Agapova, A. (2011). Conventional mutual index funds versus exchange-traded funds. *Journal of Financial Markets*, 14(2), 323-343.
- Andersen, T. G., Bollerslev, T., Diebold, F. X., & Vega, C. (2003). Micro effects of macro announcements: Real-time price discovery in foreign exchange. *American Economic Review*, 93, 38-62.
- Andersen, T. G., Bollerslev, T., Diebold, F. X., & Vega, C. (2007). Real-time price discovery in global stock, bond and foreign exchange markets. *Journal of International Economics*, 73(2), 251-277.
- Balduzzi, P., Elton, E. J., & Green, T. C. (2001). Economic news and bond prices: Evidence from the US Treasury market. *Journal of Financial and Quantitative Analysis*, 36(04), 523-543.
- Birz, G., & Lott Jr, J. R. (2011). The effect of macroeconomic news on stock returns: New evidence from newspaper coverage. *Journal of Banking & Finance*, 35(11), 2791-2800.
- Black Rock. (2010). *ETF Landscape - Global Handbook from Black Rock Q3 2010*.
- Black Rock. (2015). *ETF Landscape – Industry Highlights, June 2015*.
- Blitz, D., & Swinkels, L. (2008). Fundamental indexation: an active value strategy in disguise. *Journal of Asset Management*, 9(4), 264-269.
- Brock, W. A., & LeBaron, B. (1996). A dynamic structural model for stock return volatility and trading volume. *Review of Economics and Statistics*, 78, 94-110.
- Brenner, M., Pasquariello, P., & Subrahmanyam, M. (2009). On the volatility and comovement of US financial markets around macroeconomic news announcements. *Journal of Financial and Quantitative Analysis*, 44(06), 1265-1289.
- Charupat, N., & Miu, P. (2013). Recent developments in exchange-traded fund literature: Pricing efficiency, tracking ability, and effects on underlying securities. *Managerial Finance*, 39(5), 427-443.
- Chiang, W. (1998). *Optimizing Performance Indexing for Maximum Investment Results*. Chicago, Illinois: GPCo Publishers.
- Chung, D., & Hrazdil, K. (2013). Speed of Convergence to Market Efficiency in the ETFs Market. *Managerial Finance*, 39(5), 457-475.
- Dellva, W. (2001). Exchange-traded funds not for everyone. *Journal of Financial Planning*, 14(4), 110-124.
- Deville, L. (2008). Exchange traded funds: History, trading, and research. *Handbook of Financial Engineering* (pp. 67-98). Paris, France: Springer.
- Drenovak, M., Urošević, B., & Jelic, R. (2012). European Bond ETFs: Tracking Errors and the Sovereign Debt Crisis. *European Financial Management* 20(5), 958–994
- Elton, E. J., Gruber, M. J., Comer, G., & Li, K. (2002). Spiders: where are the bugs? *Journal of Business*, 75(3), 453-472.
- Engle, R. F., & Sarkar, D. (2006). Premiums-discounts and exchange-traded funds. *Journal of Derivatives*, 13(4), 27-45.
- Fleming, M. J., & Remolona, E. M. (1999). Price formation and liquidity in the US Treasury market: The response to public information. *Journal of Finance*, 54(5), 1901-1915.
- Fremault, A. (1991). Stock index futures and index arbitrage in a rational expectations model. *Journal of Business*, 64, 523-547.

- Frino, A., & Gallagher, D. (2001). Tracking S&P Index Funds. *Journal of Portfolio Management*, 28(1), 44-55.
- Gastineau, G. L. (2001). Exchange-traded funds: An introduction. *Journal of Portfolio Management*, 27(3), 88-96.
- Gorton, G. B., & Pennacchi, G. G. (1993). Security baskets and index-linked securities. *Journal of Business*, 66(1), 1-27.
- Green, T. C. (2004). Economic news and the impact of trading on bond prices. *Journal of Finance*, 59(3), 1201-1234.
- Guedj, I., & Huang, J. (2008). Are ETFs replacing index mutual funds: Working paper. University of Texas at Austin.
- Hamm, S. J. (2011). The effect of ETFs on stock liquidity. Available at SSRN 1687914.
- Haug, M., & Hirschey, M. (2006). The January Effect. *Financial Analysts Journal*, 62(5), 78-88.
- Hegde, S. P., & McDermott, J. B. (2004). The market liquidity of DIAMONDS, Q's, and their underlying stocks. *Journal of Banking & Finance*, 28(5), 1043-1067.
- Investment Company Institute. (2014). *Investment Company Fact Book: A Review of Trends and Activities in the U.S. Investment Company Industry* (Vol. 53). Washington, DC: Investment Company Institute.
- Jares, T. E., & Lavin, A. M. (2004). Japan and Hong Kong exchange-traded funds (ETFs): Discounts, returns, and trading strategies. *Journal of Financial Services Research*, 25(1), 57-69.
- Jensen, M. C. (1978). Some anomalous evidence regarding market efficiency. *Journal of Financial Economics*, 6(2), 95-101.
- John, K., & Lang, L. H. (1991). Insider trading around dividend announcements: theory and evidence. *Journal of Finance*, 46(4), 1361-1389.
- Jones, C. M., Lamont, O., & Lumsdaine, R. L. (1998). Macroeconomic news and bond market volatility. *Journal of Financial Economics*, 47(3), 315-337.
- Keim, D. B. (1999). An analysis of mutual fund design: the case of investing in small-cap stocks. *Journal of Financial Economics*, 51(2), 173-194.
- Kim, O., & Verrecchia, R. E. (1991). Market reaction to anticipated announcements. *Journal of Financial Economics*, 30(2), 273-309.
- Kim, O., & Verrecchia, R. E. (1997). Pre-announcement and event-period private information. *Journal of Accounting and Economics*, 24(3), 395-419.
- Kostovetsky, L. (2003). Index mutual funds and exchange-traded funds. *Journal of Portfolio Management*, 29(4), 80-92.
- Kumar, P., & Seppi, D. J. (1994). Information and index arbitrage. *Journal of Business*, 67(4), 481-509.
- Larsen, G. A., & Resnick, B. G. (1998). Empirical Insights on Indexing: How Capitalization, Stratification and Weighting Can Affect Tracking Error. *Journal of Portfolio Management*, 25(1), 51.
- Lee, C., Shleifer, A., & Thaler, R. H. (1991). Investor sentiment and the closed-end fund puzzle. *Journal of Finance*, 46(1), 75-109.
- Li, L., & Engle, R. F. (1998). *Macroeconomic announcements and volatility of treasury futures*. (Discussion Paper 98-27), Department of Economics, University of California at San Diego.
- Marshall, B. R., Nguyen, N. H., & Visaltanachoti, N. (2013). ETF arbitrage: Intraday Evidence. *Journal of Banking & Finance*, 37(9), 3486-3498.
- Nowak, S., Andritzky, J., Jobst, A., & Tamirisa, N. (2011). Macroeconomic fundamentals, price discovery, and volatility dynamics in emerging bond markets. *Journal of Banking & Finance*, 35(10), 2584-2597.

- Patro, D. K. (2001). Market segmentation and international asset prices: Evidence from the listing of World Equity Benchmark Shares. *Journal of Financial Research*, 24(1), 83-98.
- Perold, A. F. (1988). The implementation shortfall: Paper versus reality. *Journal of Portfolio Management*, 14, 4-9.
- Petersen, M. A. (2009). Estimating standard errors in finance panel data sets: Comparing approaches. *Review of Financial Studies*, 22(1), 435-480.
- Pope, P. F., & Yadav, P. K. (1994). Discovering errors in tracking error. *Journal of Portfolio Management*, 20(2), 27-32.
- Poterba, J. M., & Shoven, J. B. (2002). Exchange traded funds: A new investment option for taxable investors. *American Economic Review*, 92(2), 422-427.
- Pozen, R., & Hamacher, T. (2011). *The fund industry: How your money is managed* (Vol. 596). Hoboken, NJ: John Wiley & Sons.
- Qadan, M., & Yagil, J. (2012). On the dynamics of tracking indices by exchange traded funds in the presence of high volatility. *Managerial Finance*, 38(9), 804-832.
- Richie, N., & Madura, J. (2007). Impact of the QQQ on liquidity and risk of the underlying stocks. *Quarterly Review of Economics and Finance*, 47(3), 411-421.
- Rogers, W. (1994). Regression standard errors in clustered samples. *Stata Technical Bulletin*, 3(13).
- Roll, R. (1992). A mean/variance analysis of tracking error. *Journal of Portfolio Management*, 18(4), 13-22.
- Rompotis, G. (2005). An empirical comparing investigation on exchange traded funds and index funds performance. Available at SSRN 903110.
- Rompotis, G. (2007). Evaluating the Seasonality and Persistence of ETFs Performance and Volatility: Implications for Profitable Investing. Available at SSRN 1022876.
- Rozeff, M. S., & Kinney Jr, W. R. (1976). Capital market seasonality: The case of stock returns. *Journal of Financial Economics*, 3(4), 379-402.
- Securities Exchange Commission. (2010). Findings regarding the market events of May 6, 2010. *Report of the Staffs of the CFTC and SEC to the Joint Advisory Committee on Emerging Regulatory Issues*.
- Sharpe, W. F. (1991). The arithmetic of active management. *Financial Analysts Journal*, 40(1), 7-9.
- State Street Global Advisors. (2014). SPDR S&P 500 ETF. Retrieved 20 September 2014, from <https://www.spdrs.com/product/fund.seam?ticker=spy>
- Stoll, H. R. (1989). Inferring the Components of the Bid-Ask Spread: Theory and Empirical Tests. *Journal of Finance*, 44(1), 115-134.
- Subrahmanyam, A. (1991). A theory of trading in stock index futures. *Review of Financial Studies*, 4(1), 17-51.
- Svetina, M., & Wahal, S. (2010). Exchange traded funds: Performance and competition. *Journal of Applied Finance*, 20(2), 130.
- Tse, Y., & Martinez, V. (2007). Price discovery and informational efficiency of international iShares funds. *Global Finance Journal*, 18(1), 1-15.
- Van Ness, B. F., Van Ness, R. A., & Warr, R. S. (2005). The impact of the introduction of index securities on the underlying stocks: The case of the Diamonds and the Dow 30. *Advances in Quantitative Analysis of Finance and Accounting*, 2, 105-128.
- Vanguard Investments. (2014). ETF fees and costs. Retrieved 20 September 2014, from <https://www.vanguardinvestments.com.au/retail/ret/education/inv-library/etf-fees-and-costs.jsp>
- Wang, T., Yang, J., & Wu, J. (2006). Central bank communications and equity ETFs. *Journal of Futures Markets*, 26(10), 959-995.

Table 1: U.S. Macroeconomic News Announcements

Number	Indicator Type	Observations	Start Date ^a	Finish Date ^a	Time ^b
1	Business Inventories	88 ^c	Feb 2004	Dec 2013	10:00:00
	Consumer Confidence				
2	Index	114	May 2004	Dec 2013	10:00:00
3	Consumer Credit	109	Jun 2004	Dec 2013	15:00:00
4	Durable Goods	10 ^d	Jun 2004	Mar 2005	10:00:00
5	Target Federal Funds Rate	13 ^e	Jan 2007	Oct 2008	14:15:00
6	Factory Orders	112	Jun 2004	Dec 2013	10:00:00
7	Federal Budget Balance	112	Jan 2004	Dec 2013	14:00:00
8	New Home Sales (Units)	112	May 2004	Dec 2013	10:00:00
	Index of Leading				
9	Indicators	110	Jan 2004	Dec 2013	10:00:00
10	NAPM Index	113	Jan 2004	Dec 2013	10:00:00
	Purchasing Managers				
11	Index	111	Jun 2004	Dec 2013	10:00:00
12	Construction Spending	104	Jun 2004	Dec 2013	10:00:00

This table summarises each indicator type used in the study. Announcement and forecast data for each observation is gathered through SIRCA.

^aBeginning and end date of the announcement and expectations data sample, separated by indicator type.

^bRepresents the average release time separated by indicator type during the sample period. Target Federal Funds Rate release times fluctuate between 14:02:21 and 14:17:56. On 3rd July 2007, the Factor Orders announcement is released at 11:01:55. A Federal Budget Balance announcement is released at 12:00:01 on 11th October 2007, 14:30:01 on 14th October 2005, and 15:00:00 on 12th October 2012. On 2nd July 2007, the NAPM Index announcement and the Purchasing Managers Index announcement are released at 11.56:32.

^cSeveral Business Inventories announcements do not provide forecasts, limiting available observations.

^dDurable Goods announcement release times switched to 8:30am after March 2005.

^eNo Target Federal Funds Rate changes have been conducted since February 2010. Between October 2008, and February 2010, no forecasts were available.

Table 2: Descriptive Statistics

Variable	Mean	Standard Deviation	Minimum	Median	Maximum
TE 0, 15	5.1×10^{-4}	7.2×10^{-4}	0	2.4×10^{-4}	8.7×10^{-3}
SN	0.758	0.631	0.000	0.599	3.565
VOLA	7.6×10^{-5}	4.2×10^{-5}	3.0×10^{-5}	6.5×10^{-5}	3.8×10^{-4}
AVEQS	0.010	0.001	0.001	0.010	0.025
TDIF	2.180	2.600	3.0×10^{-4}	2.240	31.812

This table presents descriptive statistics for the SPY tracking error during the first 15 seconds after each announcement (TE 0, 15), the absolute Standardised News value for each announcement observation (SN), the standard deviation of one second returns during the 10 minutes prior to each announcement (VOLA), the average quoted bid-ask spread during the 10 minutes prior to each announcement (AVEQS), and the time between each announcement forecast and its official news release (TDIF).

Table 3: Correlation Matrix

	TE 0, 15	SN	VOLA	AVEQS	TVOL	TDIF	MANN
TE 0, 15	1.0000						
SN	0.1251	1.0000					
VOLA	0.4650	0.0917	1.0000				
AVEQS	0.2489	−0.0568	0.2874	1.0000			
TVOL	0.4084	0.0898	0.7228	−0.0433	1.0000		
TDIF	0.1118	0.0316	0.0750	−0.2261	0.2601	1.0000	
MANN	0.0903	0.0498	−0.0355	−0.0277	−0.0299	−0.0348	1.0000

This table details the correlation between the SPY tracking error during the first 15 seconds after each announcement (TE 0, 15), the absolute Standardised News value for each announcement observation (SN), the standard deviation of one second returns during the 10 minutes prior to each announcement (VOLA), the average quoted bid-ask spread during the 10 minutes prior to each announcement (AVEQS), the total SPY volume traded (in units) during the 10 minutes prior to each announcement (TVOL), the time between each announcement forecast and its official news release (TDIF), and the multiple announcements dummy variable (MANN).

Table 4: Initial Tracking Error Analysis

	1	2	3	4
SN	1.4×10^{-4} (0.000)	1.0×10^{-4} (0.001)	1.4×10^{-4} (0.000)	1.2×10^{-4} (0.000)
VOLA	No	6.992 (0.000)	No	4.395 (0.000)
AVEQS	No	0.186 (0.124)	No	0.229 (0.070)
TDIF	No	3.3×10^{-5} (0.002)	No	1.3×10^{-5} (0.144)
MANN	No	1.7×10^{-4} (0.000)	No	-3.9×10^{-6} (0.941)
INDIC F.E.?	No	No	Yes	Yes
HOUR F.E.?	No	No	Yes	Yes
DAY F.E.?	No	No	Yes	Yes
MONTH F.E.?	No	No	Yes	Yes
YEAR F.E.?	No	No	Yes	Yes
Constant	4.0×10^{-4} (0.000)	-2.1×10^{-3} (0.088)	3.1×10^{-4} (0.031)	-2.3×10^{-3} (0.081)
N	1104	1104	1104	1104
Adjusted R ²	0.0147	0.2605	0.2929	0.3797

This table presents the results from OLS estimation between 1 January 2004 and 31 December 2013. Column 1 results are based on the simple OLS regression outlined in Equation 1. Column 2 includes only control variables into the regression, while Column 3 excludes control variables and includes fixed effects. Column 4 results are based on the full model outlined in Equation 2. All regressions examine SPY tracking error during the first 15 seconds after each announcement (dependant variable) against the level of surprise generated by each announcement (SN). Column 2 includes the standard deviation of 1 second returns during the 10 minutes prior to each announcement (VOLA), the average quoted bid-ask spread during the 10 minutes prior to each announcement (AVEQS), the time between the most recent forecast and each official news release (TDIF) and a multiple announcements dummy variable (MANN). Column 3 includes an indicator type dummy variable (INDIC), the hour of the day during which each announcement was made (HOUR), the day of the week each announcement was made (DAY), the month each announcement was made (MONTH), and the year each announcement was made (YEAR) to account for fixed effects (F.E.). Column 4 includes all control variables and fixed effects. *p-values* are reported in parenthesis and are based on robust standard errors.

Table 5: Tracking Error Persistence Results

	0, 15	15, 30	30, 45	45, 60	0, 60	-15, -1	-30, -15
SN	1.2×10^{-4} (0.000)	1.7×10^{-5} (0.494)	9.9×10^{-6} (0.678)	1.3×10^{-5} (0.420)	1.0×10^{-4} (0.001)	5.8×10^{-5} (0.004)	2.9×10^{-6} (0.742)
VOLA	4.395 (0.000)	2.962 (0.000)	2.365 (0.002)	2.662 (0.000)	3.821 (0.000)	3.414 (0.000)	2.262 (0.000)
AVEQS	0.229 (0.070)	0.038 (0.553)	0.061 (0.044)	0.137 (0.007)	-0.013 (0.786)	0.049 (0.276)	1.6×10^{-3} (0.920)
TDIF	1.3×10^{-5} (0.144)	9.7×10^{-7} (0.853)	1.2×10^{-5} (0.014)	1.0×10^{-6} (0.803)	2.4×10^{-6} (0.789)	3.7×10^{-6} (0.639)	-1.2×10^{-6} (0.589)
MANN	-3.9×10^{-6} (0.941)	-1.1×10^{-5} (0.783)	1.5×10^{-6} (0.969)	2.1×10^{-5} (0.438)	-1.3×10^{-4} (0.025)	-5.3×10^{-5} (0.167)	-4.2×10^{-5} (0.042)
INDIC F.E.?	Yes	Yes	Yes	Yes	Yes	Yes	Yes
HOUR F.E.?	Yes	Yes	Yes	Yes	Yes	Yes	Yes
DAY F.E.?	Yes	Yes	Yes	Yes	Yes	Yes	Yes
MONTH F.E.?	Yes	Yes	Yes	Yes	Yes	Yes	Yes
YEAR F.E.?	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Constant	-2.3×10^{-3} (0.081)	-3.2×10^{-4} (0.632)	-9.9×10^{-4} (0.001)	-1.7×10^{-3} (0.001)	-1.4×10^{-4} (0.786)	4.2×10^{-4} (0.366)	2.3×10^{-4} (0.151)
N	1104	1104	1104	1104	1104	1104	1104
Adjusted R ²	0.3797	0.2654	0.2270	0.3523	0.3269	0.2922	0.2726

This table presents the results from OLS estimation between 1st January and 31st December 2013 with the following model outlined in Equation 2 but changes the dependant variable:

$$TE_t = \beta_0 + \beta_1 SN + \beta_2 VOLA + \beta_3 AVEQS + \beta_4 TDIF + \beta_5 CONT + \beta_6 INDIC + \beta_7 HOUR + \beta_8 DAY + \beta_9 MONTH + \beta_{10} YEAR + \varepsilon,$$

Each column regresses SPY tracking error (dependant variable) against Absolute Standardised News (SN), the standard deviation of 1 second returns during the 10 minutes prior to each announcement (VOLA), the average quoted bid-ask spread during the 10 minutes prior to each announcement (AVEQS), the time between the most recent forecast and each official news release (TDIF), a multiple announcements dummy variable (MANN), an indicator type dummy variable (INDIC), the hour of the day during which each announcement was made (HOUR), the day of the week each announcement was made (DAY), the month each announcement was made (MONTH), and the year each announcement was made (YEAR). F.E.? represents fixed effect variables. For each regression, the SPY tracking error is examined during the first 15 seconds after each announcement (0, 15), from 15 to 30 seconds after each announcement (15, 30), from 30 to 45 seconds after each announcement (30, 45), from 45 to 60 seconds after each announcement (45, 60), during the 15 seconds leading up to each announcement (-15, -1), from 30 seconds before each announcement until 15 seconds before each announcement (-30, -15), and over the first 60 seconds after each announcement (0,60). *p-values* are reported in parenthesis and are based on robust standard errors.

Table 6: Sub-Period Analysis

	1	2	3
	2004-2006	2007-2009	2010-2013
SN	3.3×10^{-5} (0.044)	1.8×10^{-4} (0.029)	1.2×10^{-4} (0.006)
VOLA	0.603 (0.107)	4.317 (0.002)	10.766 (0.000)
AVEQS	-9.3×10^{-3} (0.676)	0.344 (0.005)	-0.994 (0.008)
TDIF	-8.2×10^{-6} (0.677)	4.4×10^{-5} (0.215)	-1.8×10^{-6} (0.818)
MANN	6.9×10^{-5} (0.167)	-9.7×10^{-5} (0.474)	-7.7×10^{-6} (0.927)
INDIC F.E.?	Yes	Yes	Yes
HOUR F.E.?	Yes	Yes	Yes
DAY F.E.?	Yes	Yes	Yes
MONTH F.E.?	Yes	Yes	Yes
YEAR F.E.?	No	No	No
Constant	1.5×10^{-4} (0.531)	-3.1×10^{-3} (0.045)	9.6×10^{-3} (0.010)
N	314	349	441
Adjusted R ²	0.1476	0.3672	0.3307

This table presents the results from OLS estimation for three sub-periods using the following model outlined in Equation 2:

$$TE_t = \beta_0 + \beta_1 SN + \beta_2 VOLA + \beta_3 AVEQS + \beta_4 TDIF + \beta_5 CONT + \beta_6 INDIC + \beta_7 HOUR + \beta_8 DAY + \beta_9 MONTH + \beta_{10} YEAR + \varepsilon,$$

Each column regresses the SPY tracking error (dependant variable) against Absolute Standardised News (SN), the standard deviation of 1 second returns during the 10 minutes prior to each announcement (VOLA), the average quoted bid-ask spread during the 10 minutes prior to each announcement (AVEQS), the time between the most recent forecast and each official news release (TDIF), a multiple announcements dummy variable (MANN), an indicator type dummy variable (INDIC), the hour of the day during which each announcement was made (HOUR), the day of the week each announcement was made (DAY), and the month each announcement was made (MONTH). F.E.? represents fixed effect variables. Column 1 (2004-2006) includes all announcements released between 1st January 2004 and 31st December 2006, Column 2 (2007-2009) includes all announcements released between 1st January 2007 and 31st December 2009, and column 3 (2010-2013) includes all announcements released between 1st January 2010 and 31st December 2013. *p-values* are reported in parenthesis and are based on robust standard errors

Table 7: Robustness Check

	1	2	3	4
SN	1.2×10^{-4} (0.004)	1.2×10^{-4} (0.002)	1.6×10^{-4} (0.000)	5.1×10^{-5} (0.378)
VOLA	4.395 (0.001)	4.395 (0.000)	4.09 (0.002)	5.634 (0.003)
AVEQS	0.229 (0.024)	0.229 (0.013)	0.258 (0.044)	0.030 (0.878)
TDIF	1.3×10^{-5} (0.095)	1.3×10^{-5} (0.065)	8.9×10^{-6} (0.483)	1.6×10^{-5} (0.573)
MANN	-3.9×10^{-6} (0.959)	-3.9×10^{-6} (0.957)	Omitted	Omitted
INDIC F.E.?	Yes	Yes	Yes	Yes
HOUR F.E.?	Yes	Yes	Yes	Yes
DAY F.E.?	Yes	Yes	Yes	Yes
MONTH F.E.?	Yes	Yes	Yes	Yes
YEAR F.E.?	Yes	Yes	Yes	Yes
Constant	-2.3×10^{-3} (0.031)	-2.3×10^{-3} (0.030)	-3.0×10^{-3} (0.019)	9.9×10^{-4} (0.671)
N	1104	1104	764	340
Adjusted R ²	0.3797	0.3797	0.4128	0.3330

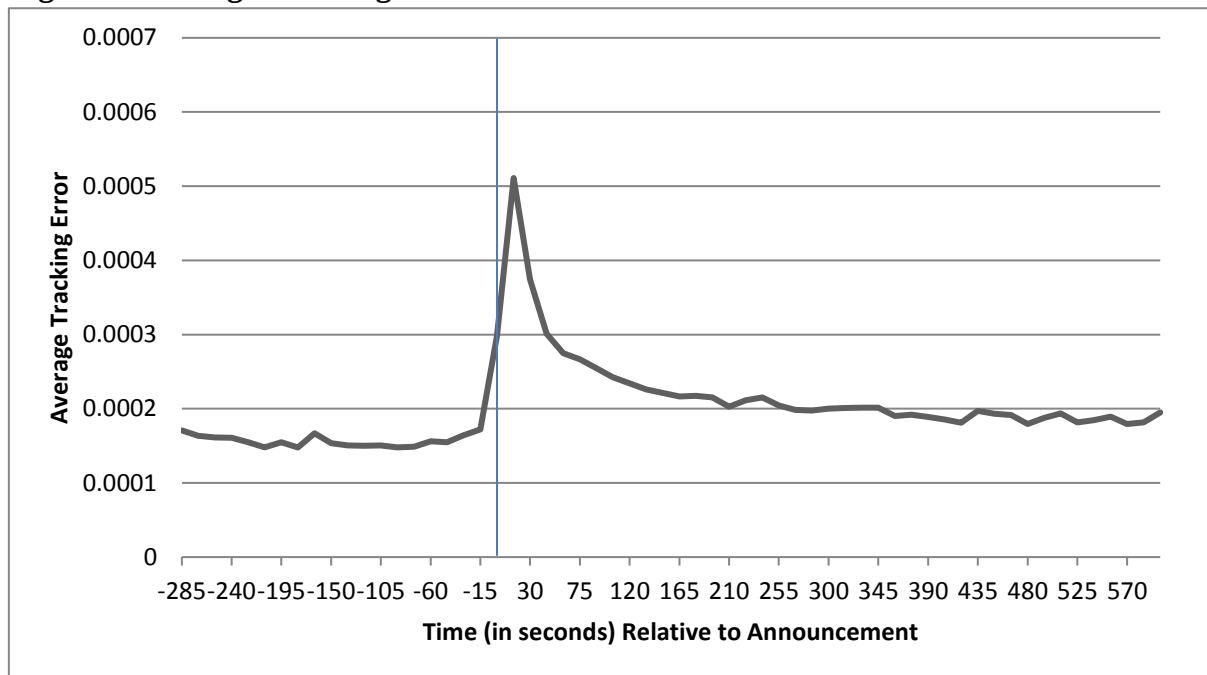
This table presents the results from OLS estimation between 1st January 2004 and 31st December 2013 using an adaption of the model outline in Equation 2:

$$TE_t = \beta_0 + \beta_1 SN + \beta_2 VOLA + \beta_3 AVEQS + \beta_4 TDIF + \beta_5 CONT + \beta_6 INDIC + \beta_7 HOUR + \beta_8 DAY + \beta_9 MONTH + \beta_{10} YEAR + \varepsilon,$$

Each column regresses the SPY tracking error (dependant variable) against Absolute Standardised News (SN), the standard deviation of 1 second returns during the 10 minutes prior to each announcement (VOLA), the average quoted bid-ask spread during the 10 minutes prior to each announcement (AVEQS), the time between the most recent forecast and each official news release (TDIF), a multiple announcements dummy variable (MANN), an indicator type dummy variable (INDIC), the hour of the day during which each announcement was made (HOUR), the day of the week each announcement was made (DAY), the month each announcement was made (MONTH), and the year each announcement was released (YEAR). F.E.? represents fixed effect variables. Column 1 clusters announcements by indicator type; Column 2 clusters announcements by indicator type and the year each announcement is released; Column 3 includes only uncontaminated announcements and omits the MANN variable; and Column 4 includes only contaminated announcements and omits the MANN variable. *p-values* are reported in parenthesis and are based on robust standard errors.

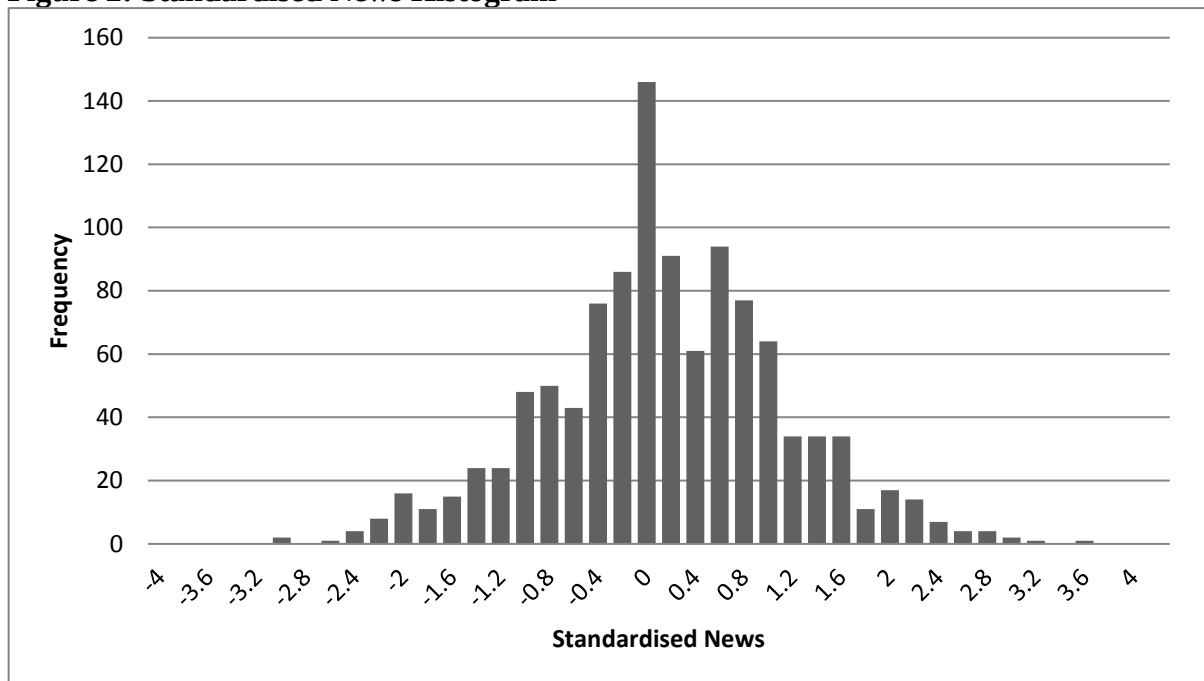
LIST OF FIGURES

Figure 1: Average Tracking Error Around Announcements



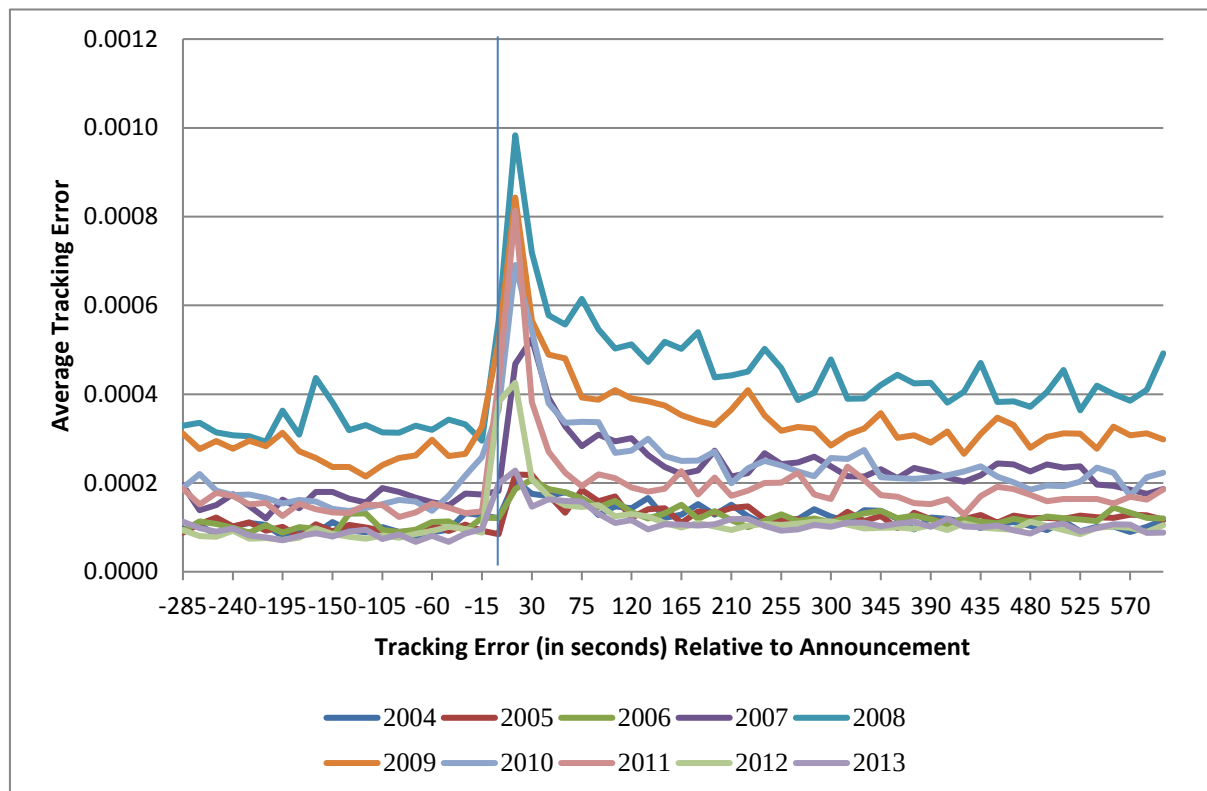
This graph details the average SPY tracking error at 15 second intervals for 5 minutes before until 10 minutes after each announcement for the time period between 1st January 2004 and 31st December 2013. Tracking error is measured as SPY and S&P 500 return deviation per 15 second interval. Each announcement is released at time zero.

Figure 2: Standardised News Histogram



This histogram outlines the frequency of raw Standardised News values for each announcement between 1st January 2004 and 31st December 2013. Each Standardised News value is placed into one bin between -4 and 4, separated by 0.20.

Figure 3: Tracking Error by Year



This graph details the average SPY tracking error each year at 15 second intervals for 5 minutes before, and 10 minutes after each announcement for the time period between 1st January 2004 and 31st December 2013. Tracking error is measured as SPY and S&P 500 return deviation per 15 second interval. Each announcement is released at time zero.